

The World Islamic Science & Education University



جامعة العلوم الإسلامية العالمية

Faculty of Information Technology

كلية تكنولوجيا المعلومات



University BookStore Database

Course Name:

Database Management System

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... Second Semester ... 2025/2026 ...



Phase 1

- University BookStore Scenario

BOOK_SALES	
SaleID	
Date	
CustomerID	
CustomerName	
CustomerPhone	
BookISBN	
BookTitle	
Author	
Publisher	
Category	
UnitPrice	
Quantity	
TotalAmount	
CashierID	
CashierName	

1.1 List all functional dependencies

Based on the Business Rules the functional dependencies:

- **SaleID** → *Date, CustomerID, CashierID*
- **BookISBN** → *BookTitle, Author, Publisher, Category, UnitPrice*
- **UnitPrice, Quantity** → *TotalAmount*
- **CashierID** → *CashierName*
- **CustomerPhone** → *CustomerID, CustomerName*



1.2 Identify all candidate keys

Based on the Functional dependences :

Step 1: Attributes Not Appearing in Right Hand Side

RHS Attributes	Attributes NOT in RHS
<i>Date, CustomerID, CashierID, BookTitle, Author, Publisher, Category, UnitPrice, TotalAmount, CashierName, CustomerName</i>	<i>SaleID, BookISBN, Quantity, CustomerPhone</i>

Step 2: Closure Calculation

(SaleID, BookISBN, Quantity, CustomerPhone)+

Start with : *(SaleID, BookISBN, Quantity, CustomerPhone)*

Using	Add
<i>SaleID → Date, CustomerID, CashierID</i>	<i>Date, CustomerID, CashierID</i>
<i>CashierID → CashierName</i>	<i>CashierName</i>
<i>BookISBN → BookTitle, Author, Publisher, Category, UnitPrice</i>	<i>BookTitle, Author, Publisher, Category, UnitPrice</i>
<i>UnitPrice, Quantity → TotalAmount</i>	<i>TotalAmount</i>
<i>CustomerPhone → CustomerID, CustomerName</i>	<i>CustomerName</i>

Result:

(SaleID, BookISBN, Quantity, CustomerPhone)+

= All Attributes

Therefore:

Candidate Key = {SaleID, BookISBN}



1.3 Normalization

1.3.1 First Normal Form (1NF)

BOOK_SALES	
<u>SaleID</u>	
Date	
CustomerID	
CustomerName	
CustomerPhone	
<u>BookISBN</u>	
BookTitle	
Author	
Publisher	
Category	
UnitPrice	
Quantity	
TotalAmount	
CashierID	
CashierName	

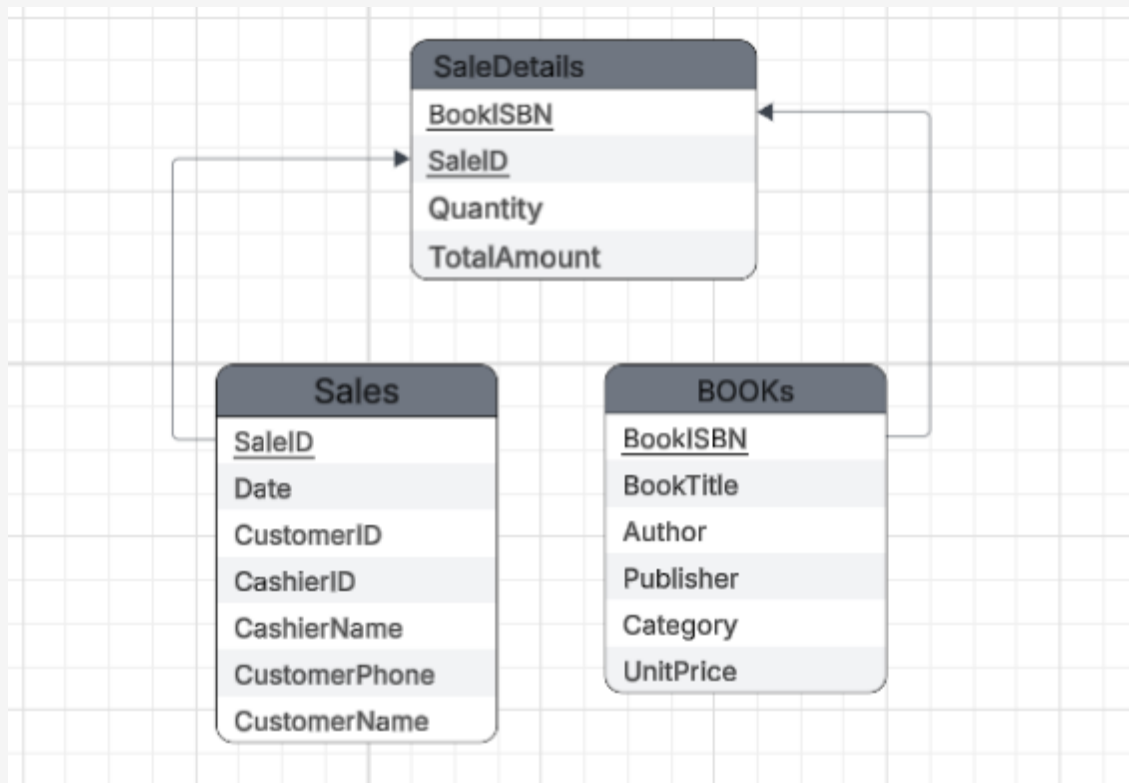
Justification:

- All attributes contain atomic values.
- No repeating groups or multi-valued attributes exist.

Therefore, the relation already satisfies 1NF



1.3.2 Second Normal Form (2NF)

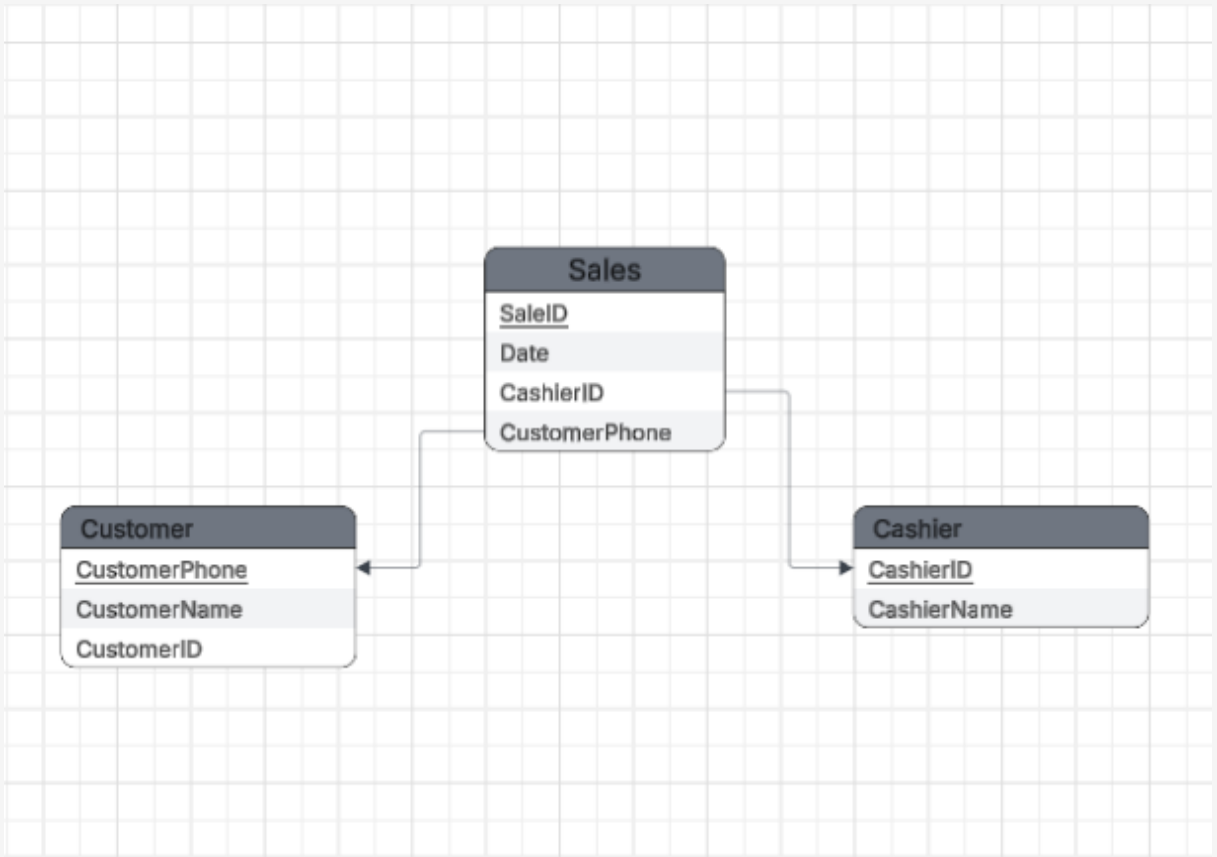


Justification:

- **Sales** : depend only on SaleID.
- **Books** : depend only on BookISBN.
- **SaleDetails** : represents the relationship between Sales and Books. Its attributes depend on the full composite key ,so it handles the many-to-many relationship



1.3.3 Third Normal Form (3NF)

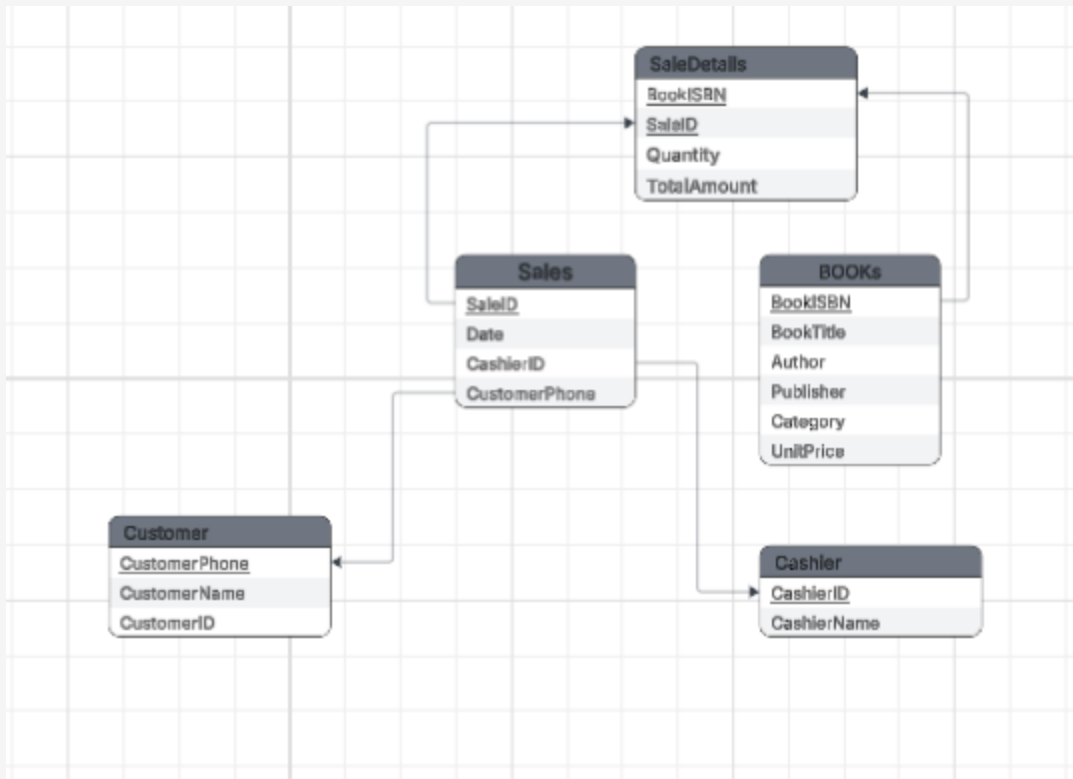


Justification:

- **Customers:** *Customer details depend on CustomerPhone.*
- **Cashiers:** *CashierName depends on CashierID.*



1.4 Final Schema



Phase 2

1.0 relational algebra queries

1.1 Selection Queries (Sigma σ)

- Query 1:** Find all books that have a price greater than 50 JD.

$$\sigma_{\text{Price} > 50}(\text{Book})$$

- Query 2:** Find all customers who live in the city of 'Amman'.

$$\sigma_{\text{City} = 'Amman'}(\text{Customer})$$



1.2 Projection Queries (π)

- Query 3:** List the titles and prices of all books in the bookstore.

$$\pi_{\text{Title,Price}}(\text{Book})$$

- Query 4:** List the names and phone numbers of all registered customers.

$$\pi_{\text{Customer_Name,Phone_Number}}(\text{Customer})$$

1.3 Join Queries ($\bowtie$)

- Query 5:** Retrieve customer names along with their order dates.

$$\pi_{\text{Customer_Name,Order_Date}}(\text{Customer} \bowtie_{\text{Customer_ID=Customer_ID}} \text{Orders})$$

- Query 6:** Display the titles of the books that have been ordered along with their specific order IDs.

$$\pi_{\text{OrderID,Title}}(\text{Order_Details} \bowtie_{\text{ISBN=ISBN}} \text{Book})$$

1.4 Set Operations ($\cup$, $-$)

- Query 7 (Union $\cup$):** Get a list of all IDs that are either authors of books or publishers in the system (Combining distinct identifier fields for a general inventory check).

$$\pi_{\text{Author_ID}}(\text{Author}) \cup \pi_{\text{Publisher_ID}}(\text{Publisher})$$

- Query 8 (Set Difference $-$):** Find the IDs of customers who have registered in the sy:

$$\pi_{\text{Customer_ID}}(\text{Customer}) - \pi_{\text{Customer_ID}}(\text{Orders})$$



1.5 Complex Expressions ()

- **Query 9 (Theta Join $\bowtie$):** Find the book titles and order IDs where the quantity ordered is greater than 3.

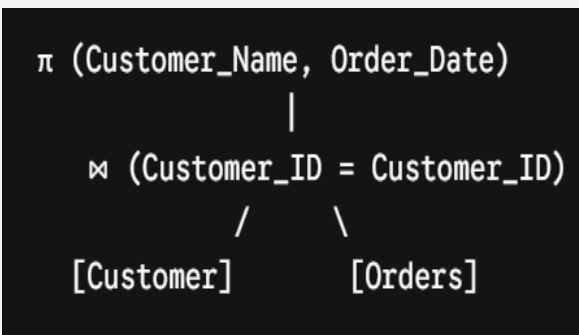
$$\pi_{\text{OrderID, Title, Quantity}}(\text{Book} \bowtie_{\text{Book.ISBN}=\text{Order_Details.ISBN} \wedge \text{Quantity} > 3} \text{Order_Details})$$

- **Query 10 (Division $\div$):** Find the IDs of customers who have purchased **every** single book available in the bookstore.

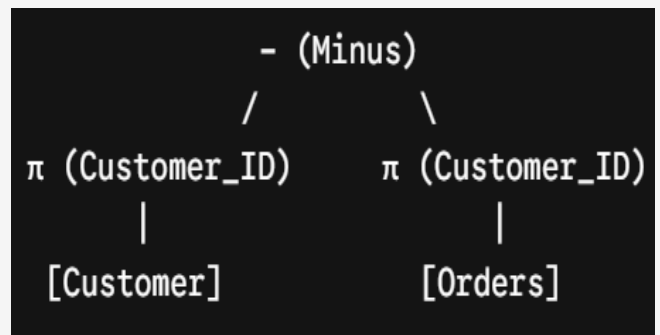
$$\pi_{\text{Customer_ID, ISBN}}(\text{Orders} \bowtie_{\text{OrderID}=\text{OrderID}} \text{Order_Details}) \div \pi_{\text{ISBN}}(\text{Book})$$

2.0 relational algebra Tree

- Tree for Query 5 (Join & Projection)



- Tree for Query 8 (Set Difference)



- Tree for Query 9 (Theta Join)

